

Organizing and Graphing Qualitative Data

Section 2.1

Course Textbook

*Introductory Statistics,
10th Edition*

Prem S. Mann

John Wiley & Sons, 2020

Based on Section 2.1

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Where Are We Going?

Today

We move from identifying variables to organizing and displaying **qualitative data**.

- ① Organize qualitative data in a frequency distribution.
- ② Calculate relative frequencies.
- ③ Represent qualitative data with a bar graph.
- ④ Represent qualitative data with a pie chart.
- ⑤ Interpret what a graph tells us.

Source: Mann (2020), *Introductory Statistics*, 10th ed., Section 2.1.

First: What Is Qualitative Data?

Recall from Chapter 1:

Qualitative / Categorical Variable

A qualitative variable classifies observations into **two or more nonnumeric categories**.

Examples

- Major: Mathematics, Biology, Computer Science, ...
- Transportation: Subway, Bus, Car, Walking
- Favorite sport: Basketball, Soccer, Baseball, Tennis

Today's question

Once we collect categorical data, how can we organize and display them?

Source: Mann (2020), *Introductory Statistics*, 10th ed., Section 2.1.

Frequency

Frequency

The **frequency** of a category is the number of observations that fall into that category.

Suppose 20 students are asked about their favorite sport:

Sport	Frequency
Basketball	9
Soccer	6
Baseball	3
Tennis	2
Total	20

Check: The frequencies must add to the total number of observations.

Source: Mann (2020), *Introductory Statistics*, 10th ed., Section 2.1.

Relative Frequency

Relative Frequency

The relative frequency tells us what proportion of the observations belong to each category.

$$\text{Relative frequency} = \frac{\text{Frequency}}{\text{Total number of observations}}$$

For Basketball:

$$\frac{9}{20} = 0.45 = 45\%$$

For Soccer: $\frac{6}{20} = 0.30 = 30\%$

Important

The relative frequencies should add to 1 (or 100%).

From Data to a Frequency Table

Favorite Sport	Frequency	Relative Frequency
Basketball	9	45%
Soccer	6	30%
Baseball	3	15%
Tennis	2	10%
Total	20	100%

Notice

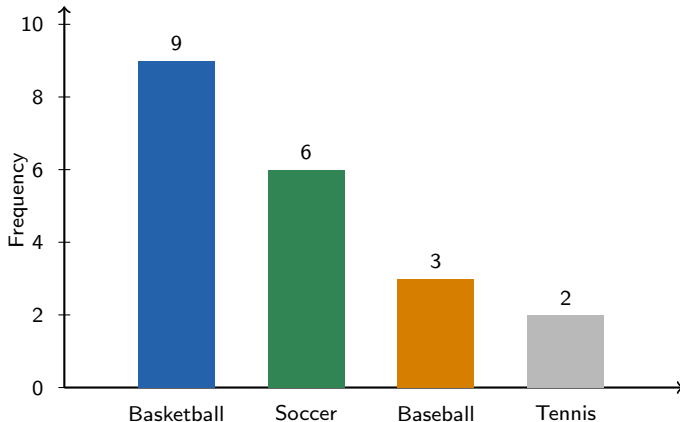
A frequency table gives us both the **counts** and the **proportions**.

Source: Mann (2020), *Introductory Statistics*, 10th ed., Section 2.1.

Bar Graph

Bar Graph

A bar graph represents the frequency or relative frequency of each category using rectangular bars.



Reading a Bar Graph

Look at the favorite-sport graph.

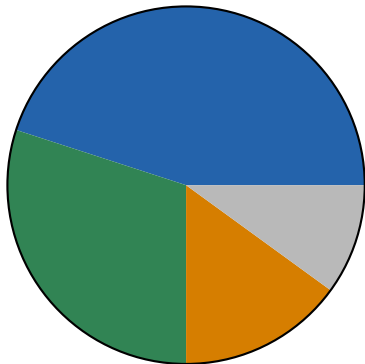
- ① Which sport is most popular?
- ② Which sport is least popular?
- ③ How many students chose Soccer?
- ④ How many more students chose Basketball than Tennis?
- ⑤ What percentage of the students chose Baseball?

Think

A graph is useful only if we can **interpret** the information it displays.

Source: Mann (2020), *Introductory Statistics*, 10th ed., Section 2.1.

Pie Chart



■ Basketball: 45% ■ Soccer: 30%
■ Baseball: 15% ■ Tennis: 10%

Pie Chart

A pie chart divides a circle into sectors. Each sector represents a proportion of the whole.

Whole = 100%

$$45 + 30 + 15 + 10 = 100\%$$

Source: Mann (2020), *Introductory Statistics*, 10th ed., Section 2.1.

Bar Graph or Pie Chart?

Bar Graph

Useful when we want to:

- compare categories
- see differences in frequency
- compare sizes easily

Pie Chart

Useful when we want to:

- emphasize parts of a whole
- show proportions or percentages
- see how categories make up 100%

Ask yourself

What is the main message I want the graph to communicate?

Source: Mann (2020), *Introductory Statistics*, 10th ed., Section 2.1.

Group Activity: Favorite Drinks

A group of 20 students was asked about their favorite drink.

Drink	Number of Students
Water	8
Coffee	5
Tea	4
Soda	3

With your group:

- 1 Find the relative frequency for each category.
- 2 Express each relative frequency as a percentage.
- 3 Check that the percentages add to 100%.
- 4 Decide whether a bar graph or pie chart would be useful.
- 5 Explain what the largest category tells us.

Source: Mann (2020), *Introductory Statistics*, 10th ed., Section 2.1.

Group Activity: Check Your Work

Drink	Frequency	Relative Frequency
Water	8	40%
Coffee	5	25%
Tea	4	20%
Soda	3	15%
Total	20	100%

For example,

$$\frac{8}{20} = 0.40 = 40\%.$$

Interpretation

Water was the most popular choice, selected by **40% of the students**.

Common Mistakes to Avoid

- 1 Forgetting to include the **total**.
- 2 Confusing **frequency** with **relative frequency**.
- 3 Forgetting to convert a decimal to a percentage when requested.
- 4 Creating percentages that do not add to approximately 100%.
- 5 Using a pie chart when the categories do not represent parts of one whole.
- 6 Forgetting to label the graph.

Quick check

Before interpreting your graph, ask:

Do the frequencies and percentages make sense?

Source: Mann (2020), *Introductory Statistics*, 10th ed., Section 2.1.

Quick Practice

A survey of 25 students asks for their preferred method of transportation to campus.

Subway	10
Bus	7
Car	5
Walking	3

Discuss with a partner:

- 1 What is the frequency of students who take the bus?
- 2 What is the relative frequency for the subway?
- 3 What percentage walk to campus?
- 4 Which category would have the largest sector in a pie chart?

Source: Mann (2020), *Introductory Statistics*, 10th ed., Section 2.1.

Exit Question

Suppose a survey produces the following frequencies:

Category A	12
Category B	5
Category C	3

- 1 What is the total number of observations?
- 2 What is the relative frequency of Category A?
- 3 What percentage of observations are in Category C?
- 4 Would a bar graph or pie chart help you communicate these results?

One sentence

What is the main difference between frequency and relative frequency?

Source: Mann (2020), *Introductory Statistics*, 10th ed., Section 2.1.

Key Takeaways

Today we learned how to organize and graph qualitative data.

- 1 **Frequency** = number of observations in a category.
- 2 **Relative frequency** = frequency divided by the total.
- 3 Relative frequencies can be written as decimals or percentages.
- 4 A **bar graph** compares categories.
- 5 A **pie chart** shows how categories make up a whole.
- 6 Always check that the frequencies and percentages make sense.

Data $\longrightarrow$ **Frequency Table** $\longrightarrow$ **Graph** $\longrightarrow$ **Interpretation**

Source: Mann (2020), *Introductory Statistics*, 10th ed., Section 2.1.

Practice for Next Class

Textbook Practice

From Mann, *Introductory Statistics*, 10th Edition:

- Exercise 2.1
- Exercise 2.5
- Exercise 2.7(b)

Next

We will move from qualitative data to **quantitative data** and learn how to organize and graph numerical data.

Source: Mann (2020), *Introductory Statistics*, 10th ed., Section 2.1.

Course Textbook

Mann, P. S. (2020).
Introductory Statistics, 10th Edition.
John Wiley & Sons.

Section 2.1

Organizing and Graphing Qualitative Data

All definitions, examples, and methods in this presentation are based on the course textbook.